Отчет по лабораторной работе № 7

Сетевые технологии

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Содержание i

1. Вывод

0.1 Цель

Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

0.2 Настройка DHCP в случае IPv4

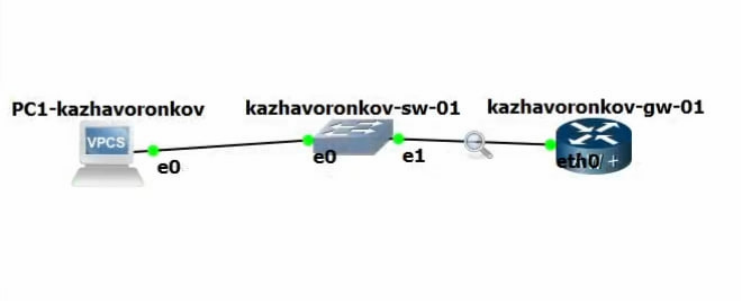

Рисунок 1: Создание сети

0.3 Настройка DHCP в случае IPv4

```
 sda    8589MB
 sdb    1MB


Install the image on? [sda]:

This will destroy all data on /dev/sda.
Continue? (Yes/No) [No]: Yes

Looking for pre-existing RAID groups...none found.
How big of a root partition should I create? (2000MB - 8589MB) [8589]MB:

Creating filesystem on /dev/sda1: OK
Done!
Mounting /dev/sda1...
What would you like to name this image? [1.5-rolling-202309110651]:
OK.  This image will be named: 1.5-rolling-202309110651
Copying squashfs image...
Copying kernel and initrd images...
Done!
I found the following configuration files:
    /opt/vyatta/etc/config/config.boot
    /opt/vyatta/etc/config.boot.default
Which one should I copy to sda? [/opt/vyatta/etc/config/config.boot]:

Copying /opt/vyatta/etc/config/config.boot to sda.
Enter password for administrator account
Enter password for user 'vyos':
Retype password for user 'vyos':
I need to install the GRUB boot loader.
I found the following drives on your system:
 sda    8589MB
 sdb    1MB


Which drive should GRUB modify the boot partition on? [sda]:

Setting up grub: OK
Done! Please reboot now.
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y
```

0.4 Настройка DHCP в случае IPv4

На маршрутизаторе изменим имя устройства и доменное имя, заменим системного пользователя.

kazhavoronkov-gw-01 - PuTTY

```
12345

  Configuration path: system login user kazahvoronkov [authentiation] is not val
id
  Set failed

[edit]
vyos@vyos# set system login user kazhavoronkov authentication plaintext-password
 12345
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - kazhavoronkov-gw0-1 ttyS0

kazhavoronkov-gw0-1 login:
```

0.5 Настройка DHCP в случае IPv4

```
 sda    8589MB
 sdb    1MB


Install the image on? [sda]:

This will destroy all data on /dev/sda.
Continue? (Yes/No) [No]: Yes

Looking for pre-existing RAID groups...none found.
How big of a root partition should I create? (2000MB - 8589MB) [8589]MB:

Creating filesystem on /dev/sda1: OK
Done!
Mounting /dev/sda1...
What would you like to name this image? [1.5-rolling-202309110651]:
OK.  This image will be named: 1.5-rolling-202309110651
Copying squashfs image...
Copying kernel and initrd images...
Done!
I found the following configuration files:
    /opt/vyatta/etc/config/config.boot
    /opt/vyatta/etc/config.boot.default
Which one should I copy to sda? [/opt/vyatta/etc/config/config.boot]:

Copying /opt/vyatta/etc/config/config.boot to sda.
Enter password for administrator account
Enter password for user 'vyos':
Retype password for user 'vyos':
I need to install the GRUB boot loader.
I found the following drives on your system:
 sda    8589MB
 sdb    1MB


Which drive should GRUB modify the boot partition on? [sda]:

Setting up grub: OK
Done! Please reboot now.
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y
```

0.6 Настройка DHCP в случае IPv4

Настроим адресацию IPv4:

kazhavoronkov-gw-01 - PuTTY

```
kazhavoronkov-gw0-1 login: kazhavoronkov
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
kazhavoronkov@kazhavoronkov-gw0-1:~$ configure
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# delete system login user vyos
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# commit
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# save
Saving configuration to '/config/config.boot'...
Done
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set interfaces ethernet eth0 address 10.0.0.1
/24
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service  dhcp-server shared-network-name
kazhavoronkov domain-name kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov name-server 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 range hosts start 10.0.0.2
```

0.7 Настройка DHCP в случае IPv4

Добавим конфигурацию DHCP-сервера на маршрутизаторе:

kazhavoronkov-gw-01 - PuTTY

```
kazhavoronkov-gw0-1 login: kazhavoronkov
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
kazhavoronkov@kazhavoronkov-gw0-1:~$ configure
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# delete system login user vyos
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# commit
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# save
Saving configuration to '/config/config.boot'...
Done
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set interfaces ethernet eth0 address 10.0.0.1
/24
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service  dhcp-server shared-network-name
kazhavoronkov domain-name kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov name-server 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 range hosts start 10.0.0.2
```

0.8 Настройка DHCP в случае IPv4

kazhavoronkov-gw-01 - PuTTY

```
kazhavoronkov-gw0-1 login: kazhavoronkov
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
kazhavoronkov@kazhavoronkov-gw0-1:~$ configure
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# delete system login user vyos
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# commit
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# save
Saving configuration to '/config/config.boot'...
Done
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set interfaces ethernet eth0 address 10.0.0.1
/24
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service  dhcp-server shared-network-name
kazhavoronkov domain-name kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov name-server 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# set service dhcp-server shared-network-name k
azhavoronkov subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
kazhavoronkov@kazhavoronkov-gw0-1# commit
```

0.9 Настройка DHCP в случае IPv4

```
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
```

Рисунок 8: Настройка PC1

0.10 Настройка DHCP в случае IPv4

```
Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
```

Рисунок 9: Настройка PC1

0.11 Настройка DHCP в случае IPv4

107	2413.611104	fe80::e00:32ff:fe06…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
108	2413.922127	fe80::e00:32ff:fe06…	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
109	2577.419222	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xcf255c68
110	2577.450458	0c:00:32:06:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
111	2578.419974	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xcf255c68
112	2578.454614	10.0.0.1	10.0.0.2	DHCP	342	DHCP Offer - Transaction ID 0xcf255c68
113	2578.480462	0c:00:32:06:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
114	2579.506524	0c:00:32:06:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
115	2581.420192	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0xcf255c68
116	2581.428735	10.0.0.1	10.0.0.2	DHCP	342	DHCP ACK - Transaction ID 0xcf255c68
117	2582.421256	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
118	2583.423011	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
119	2584.424536	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
120	2769.878273	Private_66:68:00	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.2
121	2769.885882	0c:00:32:06:00:00	Private_66:68:00	ARP	42	10.0.0.1 is at 0c:00:32:06:00:00
122	2769.887265	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xa54e, seq=1/256, ttl=64 (reply in 123)
123	2769.890801	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xa54e, seq=1/256, ttl=64 (request in 122)
124	2770.891661	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xa54e, seq=2/512, ttl=64 (reply in 125)

```
> Frame 116: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bi
> Ethernet II, Src: 0c:00:32:06:00:00 (0c:00:32:06:00:00), Dst: Private_66:68:0
> Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.2
> User Datagram Protocol, Src Port: 67, Dst Port: 68
v Dynamic Host Configuration Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xcf255c68
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 10.0.0.2
    Your (client) IP address: 10.0.0.2
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
```

```
0000  00 50 79 66 68 00 0c 00  32 06 00 00 08 00 45 10   ·Pyfh···  ····· E·
0010  01 48 00 00 00 00 80 11  25 93 0a 00 00 01 0a 00   ·H····· %······ ····
0020  00 02 00 43 00 44 01 34  73 9a 02 01 06 00 cf 25   ···C·D·4 s·····%
0030  5c 68 00 00 00 00 0a 00  00 02 0a 00 00 02 00 00   \h····· ····· ····
0040  00 00 00 00 00 00 00 50  79 66 68 00 00 00 00 00   ·····P  yfh·····
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0060  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0070  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0080  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0090  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00a0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00b0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00c0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00d0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00e0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
00f0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0100  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ····· ·· ····· ····
0110  00 00 00 00 00 00 63 82  53 63 35 01 05 36 04 0a   ····· c· Sc5··6··
0120  00 00 01 33 04 00 01 51  80 01 04 ff ff ff 00 03   ···3··· Q ····· ····
0130  04 0a 00 00 01 06 04 0a  00 00 01 0f 0c 73 76 69   ····· ·· ····· svi
0140  76 61 6e 6f 76 2e 6e 65  74 ff 00 00 00 00 00 00   vanov.ne t·····
```

Рисунок 10: Анализ пакетов

0.12 Настройка DHCP в случае IPv6

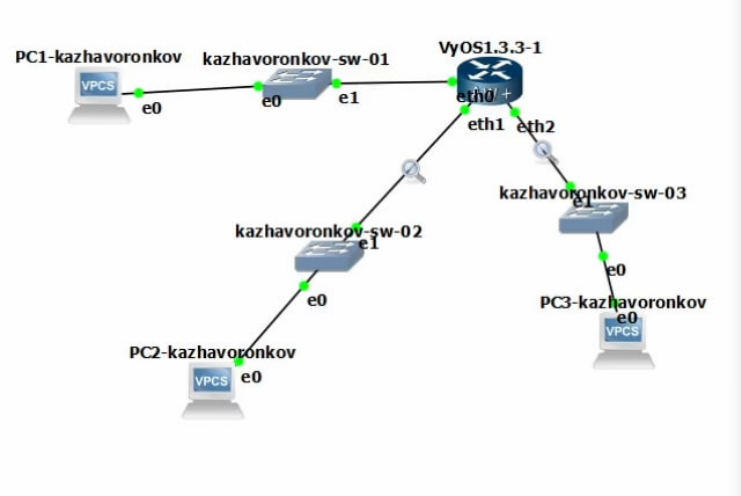

Рисунок 11: Настройка Server

0.13 Настройка DHCP в случае IPv6

VyOS1.3.3-1 - PuTTY

```
kazhavoronkov@kazhavoronkov-gw-01# set interfaces ethernet eth1 address 2000::1/
64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set interfaces ethernet eth2 address 2001::1/
64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:26:ce:01:00:00
 }
 ethernet eth1 {
+    address 2000::1/64
     hw-id 0c:26:ce:01:00:01
 }
 ethernet eth2 {
+    address 2001::1/64
     hw-id 0c:26:ce:01:00:02
 }
 }
 ethernet eth7 {
     hw-id 0c:26:ce:01:00:07
 }
 ethernet eth8 {
     hw-id 0c:26:ce:01:00:08
 }
 ethernet eth9 {
     hw-id 0c:26:ce:01:00:09
 }
 loopback lo {
 }
[edit]
kazhavoronkov@kazhavoronkov-gw-01# commit
[edit]
kazhavoronkov@kazhavoronkov-gw-01# sav
```

0.14 Настройка DHCP в случае IPv6

На маршрутизаторе настроим DHCPv6 без отслеживания состояния:

VyOS1.3.3-1 - PuTTY

```
kazhavoronkov@kazhavoronkov-gw-01# set interfaces ethernet eth1 address 2000::1/
64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set interfaces ethernet eth2 address 2001::1/
64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:26:ce:01:00:00
 }
 ethernet eth1 {
+    address 2000::1/64
     hw-id 0c:26:ce:01:00:01
 }
 ethernet eth2 {
+    address 2001::1/64
     hw-id 0c:26:ce:01:00:02
 }
 }
 ethernet eth7 {
     hw-id 0c:26:ce:01:00:07
 }
 ethernet eth8 {
     hw-id 0c:26:ce:01:00:08
 }
 ethernet eth9 {
     hw-id 0c:26:ce:01:00:09
 }
 loopback lo {
 }
```

0.15 Настройка DHCP в случае IPv6

Добавление конфигурации DHCP-сервера:

VyOS1.3.3-1 - PuTTY

```
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateless common-options domain-search kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw-01# commit
[edit]
kazhavoronkov@kazhavoronkov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address dhcp
        hw-id 0c:26:ce:01:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:26:ce:01:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:26:ce:01:00:02
    }
    ethernet eth3 {
        hw-id 0c:26:ce:01:00:03
    }
    ethernet eth4 {
        hw-id 0c:26:ce:01:00:04
    }
    ethernet eth5 {
```

0.16 Настройка DHCP в случае IPv6

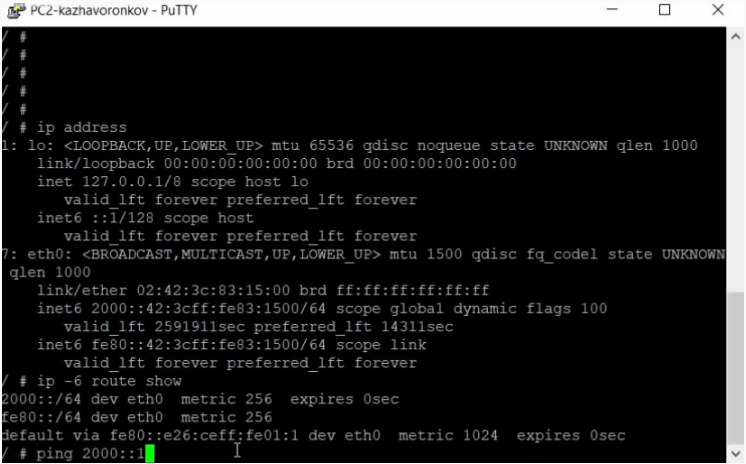

Рисунок 15: Проверка настроек PC2

0.17 Настройка DHCP в случае IPv6

```
/ # ip -6 route show
2000::/64 dev eth0  metric 256  expires 0sec
fe80::/64 dev eth0  metric 256
default via fe80::e26:ceff:fe01:1 dev eth0  metric 1024  expires 0sec
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.567 ms
64 bytes from 2000::1: seq=1 ttl=64 time=1.095 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.095/1.831/2.567 ms
/ # cat /etc/resolv.conf
/ #
```

Рисунок 16: Пооверка связи

0.18 Настройка DHCP в случае IPv6

```
/ # ping 2000::1 -c2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.059 ms
64 bytes from 2000::1: seq=1 ttl=64 time=2.270 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.059/2.164/2.270 ms
/ # cat /etc/resolv.conf
/ #
```

Рисунок 17: Проверка настроек DNS

0.19 Настройка DHCP в случае IPv6

На узле PC2 получим адрес по DHCPv6 (запрос только информации DHCPv6, но не адреса)

```
/ # ping 2000::1 -c2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.059 ms
64 bytes from 2000::1: seq=1 ttl=64 time=2.270 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.059/2.164/2.270 ms
/ # cat /etc/resolv.conf
/ #
```

Рисунок 18: Получение адреса

0.20 Настройка DHCP в случае IPv6

Пропингуем от PC2 маршрутизатор

```
/ # ip -6 route show
2000::/64 dev eth0  metric 256  expires 0sec
fe80::/64 dev eth0  metric 256
default via fe80::e26:ceff:fe01:1 dev eth0  metric 1024  expires 0sec
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.567 ms
64 bytes from 2000::1: seq=1 ttl=64 time=1.095 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.095/1.831/2.567 ms
/ # cat /etc/resolv.conf
/ #
```

Рисунок 19: Проверка связи

0.21 Настройка DHCP в случае IPv6

Проверим настройки DNS

```
/ # ping 2000::1 -c2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.059 ms
64 bytes from 2000::1: seq=1 ttl=64 time=2.270 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.059/2.164/2.270 ms
/ # cat /etc/resolv.conf
/ #
```

Рисунок 20: Проверка DNS

0.22 Настройка DHCP в случае IPv6

```
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
```

Рисунок 21: Статистика и выданные адреса

0.23 Настройка DHCP в случае IPv6

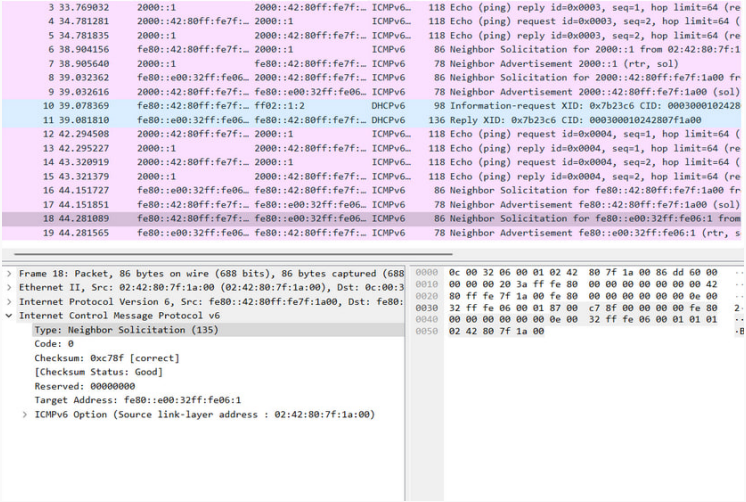

Рисунок 22: Захваченый трафик и анализ информации

0.24 Настройка DHCP в случае IPv6

На маршрутизаторе настроим DHCPv6 с отслеживанием состояния

VyOS1.3.3-1 - PuTTY

```
[1]+  Stopped                 /opt/vyatta/bin/vyatta-op-cmd-wrapper "$@"
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service router-advert interface eth2 mana
ged-flag
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 name-server 2001::1
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 domain-search kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 address-range start 2001::100 stop 200
1::199
[edit]
kazhavoronkov@kazhavoronkov-gw-01# commit
[edit]
```

0.25 Настройка DHCP в случае IPv6

Добавим конфигурацию DHCP-сервера на маршрутизаторе:

VyOS1.3.3-1 - PuTTY

```
[1]+  Stopped                 /opt/vyatta/bin/vyatta-op-cmd-wrapper "$@"
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service router-advert interface eth2 mana
ged-flag
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 name-server 2001::1
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 domain-search kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 address-range start 2001::100 stop 200
1::199
[edit]
kazhavoronkov@kazhavoronkov-gw-01# commit
[edit]
```

0.26 Настройка DHCP в случае IPv6

На маршрутизаторе посмотрим выданные адреса:

```
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
kazhavoronkov@kazhavoronkov-gw-01#
```

Рисунок 25: Просмотр выданных адресов

0.27 Настройка DHCP в случае IPv6

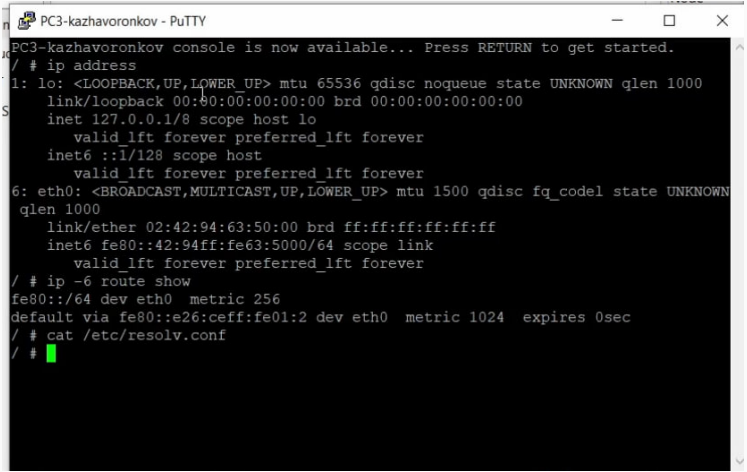

Рисунок 26: Настройки PC3

0.28 Настройка DHCP в случае IPv6

На узле PC3 проверим настройки DNS:

```
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
default via fe80::e26:ceff:fe01:2 dev eth0  metric 1024  expires 0sec
/ # cat /etc/resolv.conf
/ # udhcpv6 -i eth0
/bin/sh: udhcpv6: not found
/ # udhcpc6 -i eth0
udhcpc6: started, v1.37.0
udhcpc6: sending discover
udhcpc6: sending select
udhcpc6: IPv6 obtained, lease time 43200
/ #
```

Рисунок 27: Настройки DNS

0.29 Настройка DHCP в случае IPv6

На узле PC3 получим адрес по DHCPv6:

PC3-kazhavoronkov - PuTTY

```
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN
 qlen 1000
    link/ether 02:42:94:63:50:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:94ff:fe63:5000/64 scope link
       valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
default via fe80::e26:ceff:fe01:2 dev eth0  metric 1024  expires 0sec
/ # ping 2001::1 -c 2
PING 2001::1 (2001::1): 56 data bytes
64 bytes from 2001::1: seq=0 ttl=64 time=2.800 ms
64 bytes from 2001::1: seq=1 ttl=64 time=0.943 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.943/1.871/2.800 ms
/ # cat /etc/resolv.conf
search kazhavoronkov.net
nameserver 2001:0000:0000:0000:0000:0000:0000:0001
/ #
```


Рисунок 28: Получение адреса

0.30 Настройка DHCP в случае IPv6

VyOS1.3.3-1 - PuTTY

```
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 domain-search kazhavoronkov.net
[edit]
kazhavoronkov@kazhavoronkov-gw-01# set service dhcpv6-server shared-network-name
 kazhavoronkov-stateful subnet 2001::0/64 address-range start 2001::100 stop 200
1::199
[edit]
kazhavoronkov@kazhavoronkov-gw-01# commit
[edit]
kazhavoronkov@kazhavoronkov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
kazhavoronkov@kazhavoronkov-gw-01# run sow dhcpv6 server leases

  Invalid command:  [sow]

[edit]
kazhavoronkov@kazhavoronkov-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration     Remaining
 Type           Pool                    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
 -------------  ----------------------  ------------------------------------
-
2001::198       active   2025/12/20 11:11:56   2025/12/20 23:11:56  11:58:49
 non-temporary  kazhavoronkov-stateful  6d:74:58:12:00:03:00:01:02:42:94:63:50:0
0
[edit]
kazhavoronkov@kazhavoronkov-gw-01#
```

0.31 Настройка DHCP в случае IPv6

```
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
```

Рисунок 30: Пинг и проверка DNS

0.32 Настройка DHCP в случае IPv6

```
Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
```

Рисунок 31: Выданные адреса

0.33 Настройка DHCP в случае IPv6

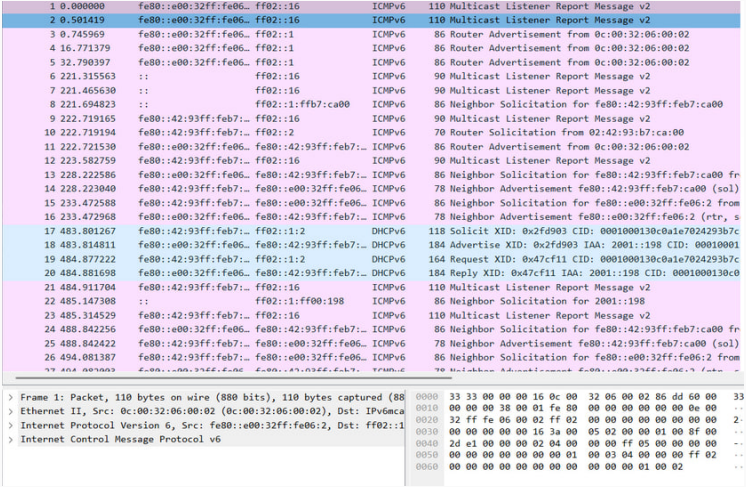

Рисунок 32: Захваченный трафик

1. Вывод

1.1 Вывод

В ходе выполнения лабораторной работы мы получили навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.